



Exploratory Data Analysis
Lab Activity 2: Visualizing & modeling relationships

Instruction

1. Start a new RMarkdown document.
2. Load the following packages at the top of your Rmd: dplyr, ggplot2, fivethirtyeight
3. When interpreting visualizations, models, etc, be sure to do so in a contextually meaningful way.
4. This homework is a resource for *you*. Record all work that is useful for your current learning & future reference. Further, try your best, but don't stay up all night trying to finish all of the exercises! We'll discuss any questions / material you didn't get to next meeting.

Goals:

The goal of this homework is to get extra practice with visualizing relationships, constructing models, and interpreting models. You'll also explore three new ideas:

- interactions between predictors
- controlling for covariates
- residuals & least squares estimation

Activity:

In their research into the "campaign value of incumbency," Benoit and Marsh (2008) - http://www.kenbenoit.net/pdfs/ajps_348.pdf - collected data on the campaign spending of 464 candidates for the 2002 elections to the Irish Dail. The authors provide the following data set

```
campaigns <- read.csv("https://www.macalester.edu/~ajohns24/data/CampaignSpending.csv") %>%
  select(votes1st, incumb, totalexp) %>%
  na.omit()
```

which, for all 464 candidates, measures many variables including

variable	meaning
votes1st	number of "1st preference" votes the candidate received
incumb	No if the candidate was a challenger, Yes if the candidate was an incumbent
totalexp	number of Euros spent by the candidate's campaign

1. Votes vs Incumbency

We might be able to explain some of the variability in votes received by a candidate's incumbency status.

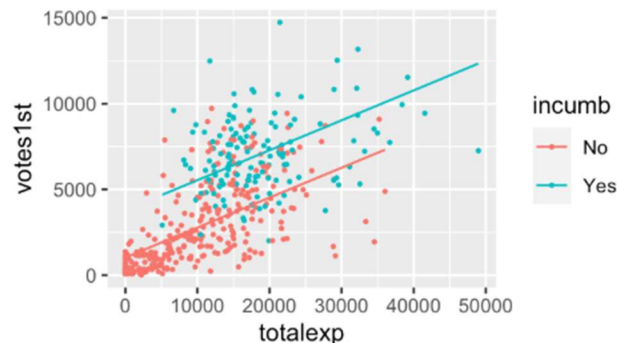
- a. Construct a visualization of the relationship between votes1st and incumb.



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- b. Construct a model of votes1st by incumb, write out the model formula, & interpret all coefficients.
 - c. From the model coefficients, compute the average votes received by incumbents and the average votes received by challengers. We'll learn another method to do these computations tomorrow.
2. **Votes vs Incumbency Status & Campaign spending**
Let's incorporate campaign spending into our analysis.
 - a. Construct 1 visualization of the relationship of votes1st vs incumb and totalexp.
 - b. Construct a model of votes1st by incumb and totalexp. Store this as model2. Write out the following formulas:
 1. the full model formula
 2. a simplified formula for challengers
 3. a simplified formula for incumbents
 - c. Interpret all coefficients. Remember: your interpretation of the incumb coefficient here should be different than in the first model since the models contain different sets of predictors.
 - d. Use this model to predict the number of votes received by the following candidates.
 1. Candidate 1: a challenger that spends 10,000 Euros
 2. Candidate 2: an incumbent that spends 10,000 Euros
 - e. Check your predictions using the predict() function.
3. Check out a plot of model2. (RStudio code is included FYI, but don't worry about it for now!)

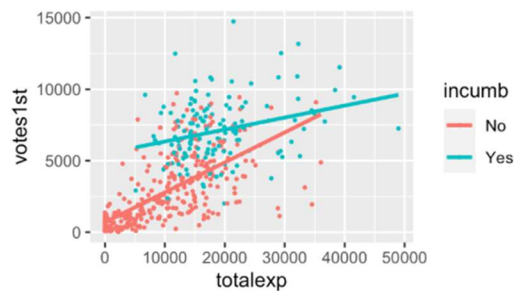
```
ggplot(campaigns, aes(x = totalexp, y = votes1st, color = incumb)) +  
  geom_point(size = 0.5) +  
  geom_line(aes(y = model2$fitted.values))
```





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Since the lines are parallel, this model assumes that incumbents and challengers enjoy the same return on campaign spending. However, notice from the plot that this may not be the best assumption. Rather, without the parallel model constraint, the trend looks more like this (you'll make this plot later):



To allow our models for challengers and incumbents to have different intercepts *and* different slopes, we can type `totalexp * incumb` instead of `totalexp + incumb` in the `lm` function:

```
new_model <- lm(votes1st ~ totalexp * incumb, campaigns)
summary(new_model)
```

- The `totalexp : incumbYes` term that you see in the model summary is called an **interaction term**. Mathematically, it's the *product* of these two variables, `totalexp * incumbYes`. With this in mind, write down the model formulas:
 - the full model formula (of the form $\text{votes1st} = a + b \text{ totalexp} + c \text{ incumbYes} + d \text{ totalexp} * \text{incumbYes}$)
 - a simplified formula for challengers (of the form $\text{votes1st} = a + b \text{ totalexp}$)
 - a simplified formula for incumbents (of the form $\text{votes1st} = a + b \text{ totalexp}$)
- Use this model to predict the number of votes received by the following candidates. Calculate these by hand & then check your work using `predict()`.
 - Candidate 1: a challenger that spends 10,000 Euros
 - Candidate 2: an incumbent that spends 10,000 Euros
- You can visualize this model by adding a `geom_smooth()` to your scatterplot:

```
ggplot(campaigns, aes(x = totalexp, y = votes1st, col = incumb)) +
  geom_point() +
  geom_smooth(method = "lm")
```



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You should notice the **interaction** between campaign spending & incumbency status, i.e. that the relationship between votes & spending differs for incumbents and challengers. With this in mind, comment on what the differing intercepts and differing slopes indicate about the relationship between the three variables of interest.

- d. Putting this all together, interpret all four model coefficients from `new_model`. Hint: As we have for models in the past, look back to the equations for the incumbent & challenger models.

4. Correlation does not imply causation

If you're single, this model probably didn't inspire you to go out and find a spouse. Why? Just because there's a relationship between wages and marital status doesn't mean being single *causes* a person to earn less. List 10 *confounding variables* that might explain this relationship between wages and marital status.

5. Including covariates / confounding variables

One variable that might explain the observed relationship between wages & marital status is age - older people both tend to make more money and are more likely to be married. We can control for this covariate by including it in our model.

- a. Fit and summarize a model of wage by marital and age that does *not* include an interaction term.
- b. Construct a visualization of this relationship that eliminates individual data points but includes the model smooth or trend. Does it appear that age and marital status interact? Explain.
- c. Compare two workers that are both 20 years old, but one is married and the other is single. By how much do their predicted wages differ? Use the model formula to calculate this difference and the plot to provide intuition.
- d. Compare two workers that are both 30 years old but one is married and the other is single. By how much do their predicted wages differ?

6. Controlling for more covariates

Taking age into account added some insight into the discrepancy between single and married workers' wages. Let's see what happens when we control for even more variables. To this end, model a person's wage by their marital status while controlling for their age, years of education, and the job industry in which they work. For simplicity, eliminate all interaction terms:

```
cps_mod3 <- ____  
summary(cps_mod3)
```



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- a. This is difficult model to visualize since there are 2 quantitative predictors (age and education) and 2 categorical predictors (marital and industry) with a possible 12 category combinations (2 marital statuses * 6 industries). If you *had* to draw it, what would it look like?
 - 12 parallel lines
 - 12 non-parallel lines
 - 12 parallel planes
 - 12 non-parallel planes
 - b. Compare two workers that are both 20 years old, have 16 years of education, and work in the service industry. If one is married and the other is single, by how much do their predicted wages to differ?
 - c. Compare two workers that are both 30 years old, have 12 years of education, and work in the construction industry. If one is married and the other is single, by how much do their predicted wages to differ?
 - d. In light of b & c, interpret the maritalsingle coefficient.
 - e. In conclusion, we saw different maritalsingle coefficients in each of `cps_mod1`, `cps_mod2`, and `cps_mod3`. Explain the significance of the difference between these measurements - what insight does it provide?
7. Answer the following questions related to the coefficients in `cps_mod3`.
- a. What is the reference level of the industry variable?
 - b. For fixed education, marital status, and age, in what industry do workers make the most money? The least?
 - c. Interpret the education coefficient.
 - d. Interpret the industrymanagement coefficient.